INTERIM FEDERAL SPECIFICATION

PULLER, MECHANICAL, WATCH AND CLOCK HAND

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20407, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope.- This specification covers one type of hand tool that is used by jewelers for removing the hand(s) from watches and small clocks.

1.1.1 Federal specification coverage.- Federal specifications do not cover all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.- The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:
Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 151 - Metals; Test Methods.

FOR INSTRUCTIONS ON OBTAINING COPIES OF FEDERAL SPECIFICATIONS AND STANDARDS, SEE LAST PAGE.

Military Specification:
MIL-H-15424 - Hand Tools; Packaging of.

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Marking.- Each puller shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.2 Finish.- Except for two small metal links in the compound leverage toggle mechanism, the exposed metal surfaces of the tool shall be plated with nickel, chromium, or nickel and chromium; unless the metal is corrosion resisting steel in which case no plating will be required. The two small metal links shall be phosphate coated, oxidized coated, blued steel; or any of the platings or corrosion resisting material specified in the preceding sentence will be acceptable.

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3.3 Material.- The toggle mechanism, actuating knob, and ferrules, if used, shall be of metal. The tool handle shall be made of either metal or plastic.

3.4 Design and construction.- The tool shall be designed and constructed to remove watch hands and small clock hands, and retain the clock hand in the tool until purposely released. The tool shall be so designed that when the operator presses the actuating knob at the end of the handle, the push rod will slide through the handle and actuate a toggle mechanism, which in turn shall open up the jaws so that the tool can be placed over the clock hand. Then by reversing the direction, or pulling on the actuating knob, the jaws of the tool close up under the watch hand and removes the hand from the clock or watch. A spring operated plunger shall be provided in the jaw end of the tool for the purpose of holding the clock hand in the tool until purposely released. The periphery of the actuating knob shall be knurled or serrated.

3.4.1 Spread range.- The spread range shall be at least 7/16-inch, measured between the widest opening between the concaved surfaces of the jaws when the puller is fully opened.

3.4.2 Reach.- The reach shall be at least 1/8-inch, measured from a straight line at the widest opening in the concaved surfaces of the jaws to the top of the clock hand retaining plunger when it is fully compressed.

3.4.3 Dimensional requirements.- The overal length of the tool shall be not less than four inches. The jaw width shall be 1/8-inch, plus or minus 1/32-inch. The overall dimensions of the handle shall be not less than 5/16-inch diameter and not less than 1-1/2 inches in length. The actuating knob shall be not less than 1/4-inch in diameter and not less than 1/8-inch in length on the gripping portion.

3.5 Hardness.- The two jaws shall be hardened to not less than 54 nor more than 62 Rockwell C (see 4.4).

3.6 Workmanship.- Workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.

4.2.1 Inspection lot.- All pullers presented at one time shall be considered a lot for purposes of quality conformance inspection.

4.2.2 Sampling pullers for examination.- A random sample of pullers shall be selected from each lot for examination in accordance with MIL-STD-105 at inspection level II. The acceptable quality level (AQL) shall be equal to 1.5 percent defective.

4.2.3 Sampling for hardness test.- A random sample of pullers shall be selected from each lot for test in accordance with MIL-STD-105 at inspection level S-4. The acceptable quality level (AQL) shall be equal to 4.0 percent defective.

4.3 Inspection.

4.3.1 Examination.- Each of the sample pullers selected in accordance with 4.2.2 shall be examined to verify conformance with this specification. Examination shall be conducted as specified in table I. Any puller in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective pullers in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

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Table I. - Classification of puller defects in accordance with MIL-STD-105

Categories		Defects
Critical:		
1	----------	None defined.
Major:		
101	----------	Finish, not as specified.
102	----------	Material, not as specified.
103	----------	Design and construction, not as specified.
104	----------	Spread range, not at least 7/16-inch.
105	----------	Reach, not at least 1/8-inch.
106	----------	Dimensional requirements, not as specified.
107	----------	Workmanship, not as specified.
Minor:		
201	----------	Marking, manufacturer's name or trademark missing, incorrect, illegible or not permanent.
202	----------	Preservation, packaging, and packing not as required.

4.4 Hardness test.- The Rockwell hardness test shall be conducted in accordance with method 243.1 of Fed. Test Method Std. No. 151 to determine conformance with 3.5.

4.5 Inspection of preparation for delivery.- The preservation, packaging, packing, and marking of the pullers shall be examined to determine compliance with the requirements of Military Specification MIL-H-15424 for Dividers, Drafting, Plain, Bow.

5. PREPARATION FOR DELIVERY
(Civil agencies. Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment).

5.1 Cleaning, preservation and packaging.

5.1.1 Level A.- Preservation and packaging of pullers shall be in accordance with level A of MIL-H-15424 for Dividers, Drafting, Plain, Bow.

5.1.2 Level C.- Cleaning, preservation, and packaging shall be in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 Level A.- Pullers packaged as specified (see 6.2), shall be packed in accordance with level A requirements of MIL-H-15424 for Dividers, Drafting, Plain, Bow.

5.2.2 Level B.- Pullers packaged as specified (see 6.2), shall be packed in accordance with level B requirements of MIL-H-15424 for Dividers, Drafting, Plain, Bow.

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5.2.3 Level C.- Pullers packaged as specified (see 6.2), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Military.- In addition to any special marking required by the contract or order, or herein, interior packages and exterior shipping containers shall be marked in accordance with MIL-H-15424.

5.3.2 Civil agencies. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Ordering data.- Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Level of packaging and packing required (see 5.1 and 5.2).

(c) Marking, if different (see 5.3).

Notice.- SEE LAST PAGE .

GSA DC 84-11911

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